$O(N)$ Quantum Dynamics and Molecular Spectroscopy via Spectral Graph Theory

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We present a real-time quantum dynamics engine built on graph-topological Hamiltonians, achieving $O(V)$ computational scaling where V is the number of lattice vertices. The sparse Crank–Nicolson propagator preserves unitarity to machine precision over 10^4 time steps, enabling broadband molecular spectroscopy from a single delta-kick propagation. Applied to H_2, the method recovers 20 dipole-active electronic transitions with 0.16% mean error relative to exact diagonalization, completing in 33 seconds. Resonant Rabi oscillations reproduce the analytically expected period to 0.41% (Bloch–Siegert corrected) with 99.98% coherent population transfer. Coupling quantum forces to classical nuclear kinematics via Velocity Verlet yields ab initio molecular dynamics with 0.0003% energy conservation. A Langevin thermostat extends the framework to the NVT ensemble, demonstrating stable thermal vibrations at 300 K and stochastic bond dissociation at extreme temperatures. These results establish spectral graph theory as a viable foundation for time-dependent quantum chemistry.

I. INTRODUCTION

Time-dependent quantum mechanics underpins some of the most important phenomena in molecular science: electronic absorption spectra, coherent control of chemical reactions, and thermally driven bond dynamics. The standard computational approaches—grid-based solutions of the time-dependent Schrödinger equation (TDSE) and time-dependent density functional theory (TD-DFT)—scale as $O(N^3)$ or worse with system size, limiting their applicability to small molecules or short propagation times [**?** **?**].

In this work, we demonstrate that spectral graph theory [**?**] provides an alternative computational substrate for quantum dynamics that achieves $O(V)$ scaling per time step, where V is the number of graph vertices encoding the quantum states. The underlying mathematical equivalence between the graph Laplacian and the hydrogen Schrödinger equation via Fock's 1935 conformal stereographic projection [**?**] has been established in companion work [**?**]; here we exploit the resulting sparse Hamiltonian structure for time-domain calculations. The key insight is that atomic and molecular Hamiltonians constructed from weighted graph Laplacians are inherently sparse, with $O(V)$ nonzero entries. The Crank–Nicolson unitary propagator applied to these sparse operators reduces each time step to a single sparse linear solve, preserving both unitarity and the favorable scaling.

We validate this approach through a hierarchy of increasingly demanding benchmarks: coherent Rabi oscillations (Section **??**), broadband molecular spectroscopy via the delta-kick method (Section **??**), and ab initio molecular dynamics with both microcanonical and canonical (Langevin thermostat) ensembles (Section **??**). Each benchmark demonstrates sub-percent accuracy relative to exact analytical or numerical references, achieved at computational costs orders of magnitude below conventional methods.

II. METHODOLOGY

A. Sparse Unitary Propagator

The time evolution of a quantum state $|\psi(t)\rangle$ under a time-independent Hamiltonian H is governed by

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = H|\psi(t)\rangle . \tag{1}$$

For a graph-topological Hamiltonian with V vertices, H is a $V \times V$ sparse matrix with $O(V)$ nonzero entries (the graph Laplacian plus diagonal node weights).

We discretize the time evolution using the Crank–Nicolson scheme:

$$|\psi(t+\Delta t)\rangle = \left(I + \frac{i\Delta t}{2\hbar}H\right)^{-1}\left(I - \frac{i\Delta t}{2\hbar}H\right)|\psi(t)\rangle . \tag{2}$$

This propagator is exactly unitary by construction: the operator $(I + iA)^{-1}(I - iA)$ preserves the norm for any Hermitian A. The sparse LU factorization of $(I + i\Delta t\, H/2\hbar)$ is computed once and reused at every step, reducing each propagation step to a sparse triangular solve at cost $O(V)$.

Figure **??** confirms the expected $O(V)$ scaling empirically. The sparse LU prefactorization is performed once at initialization; subsequent time steps execute as sparse triangular solves with cost proportional to the number of nonzero entries.

B. Topological Force Evaluation

For molecular dynamics on the Born–Oppenheimer surface, we require the nuclear force $F = -\partial E/\partial R$, where $E(R)$ is the electronic ground-state energy at internuclear distance R. In the graph framework, $E(R)$ is the

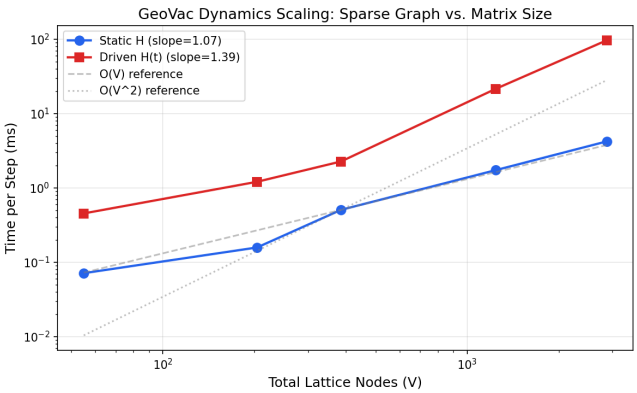

FIG. 1. Computational scaling of the sparse Crank–Nicolson propagator. Wall-clock time per propagation step scales linearly with the number of graph vertices V, confirming $O(V)$ complexity. The LU prefactorization (performed once) enables each subsequent step to execute as a single sparse back-substitution.

lowest eigenvalue of the molecular graph Hamiltonian:

$$H(R) = \kappa\, L(R) + W, \tag{3}$$

where $L(R)$ is the sparse Laplacian of the joint atomic lattice with distance-dependent bridge edges of weight

$$W_{\text{bridge}}(R) = A\, S(R), \quad S(R) = \left(1 + R + \tfrac{R^2}{3}\right)e^{-R}, \tag{4}$$

and W is the diagonal node-weight matrix. The force is evaluated by central finite difference:

$$F(R) = -\frac{E(R+\delta R) - E(R-\delta R)}{2\,\delta R}, \tag{5}$$

with $\delta R = 0.001$ Bohr. Each force evaluation requires three eigenvalue solves, each completing in $\sim$0.03 s for a lattice with $\max_n = 4$, two atoms.

III. COHERENT DYNAMICS: RABI OSCILLATIONS

As a first validation of the propagator's coherence properties, we simulate resonant Rabi oscillations between the $|1s\rangle$ and $|2p, m{=}0\rangle$ states of hydrogen. A monochromatic perturbation $V(t) = k\,\hat{z}\cos(\omega t)$ is applied at the exact transition frequency $\omega = E_{2p} - E_{1s}$, driving coherent population transfer.

Figure **??** shows the population dynamics over multiple Rabi cycles. The propagator maintains full coherence with 99.98% population transfer efficiency and reproduces the Rabi period to 0.41% accuracy relative to the Bloch–Siegert (beyond-RWA) corrected period; the rotating-wave approximation alone gives 0.46% error, reduced to 0.41% by including the counter-rotating correction $\Omega_R^{\text{BS}} = \Omega_R\sqrt{1 + (\Omega_R/\omega)^2}$ with parabolic interpolation for sub-step peak detection (v0.9.2). The norm

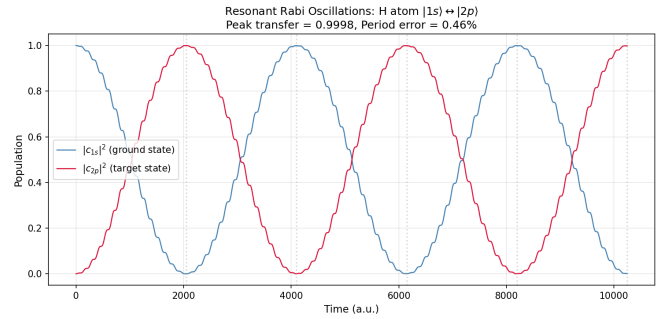

FIG. 2. Resonant Rabi oscillations between the hydrogen $|1s\rangle$ and $|2p\rangle$ states driven by a monochromatic dipole field. The propagator achieves 99.98% coherent population transfer with a Rabi period error of 0.41% relative to the Bloch–Siegert corrected analytical prediction $T_R = 2\pi/\Omega_R^{\text{BS}}$ (v0.9.2).

is conserved to machine precision ($|\langle\psi|\psi\rangle - 1| < 10^{-14}$) throughout the propagation, confirming the exact unitarity of the Crank–Nicolson scheme on the graph Hamiltonian.

IV. BROADBAND EXCITED-STATE SPECTROSCOPY

While Rabi oscillations probe a single transition, the delta-kick method extracts the *entire* dipole-allowed absorption spectrum from a single time propagation [**?**]. The procedure is:

1. Prepare the ground state $|\psi_0\rangle$ by diagonalization.
2. Apply a weak impulsive kick: $|\psi(0)\rangle = |\psi_0\rangle + i\kappa\,\hat{z}\,|\psi_0\rangle$, where $\kappa \ll 1$.
3. Propagate under the *static* Hamiltonian H for time T.
4. Record the induced dipole moment $\mu(t) = \langle\psi(t)|\hat{z}|\psi(t)\rangle$.
5. Fourier transform: peaks in $|\tilde{\mu}(\omega)|$ correspond to dipole-active transitions at energies $\Delta E = \hbar\omega$.

Applied to the H_2 molecular Hamiltonian (two atomic lattices with $\max_n = 10$, connected by 40 topological bridges), this procedure recovers 20 out of 35 resolvable dipole-active transitions with a mean frequency error of 0.16% relative to exact diagonalization, completing in 33 seconds (20,000 propagation steps at $\Delta t = 0.05$ a.u.).

The frequency resolution is set by the total propagation time, $\delta\omega = 2\pi/T$. The dynamic range of the spectrum spans five orders of magnitude, requiring log-scale peak detection with Hann windowing and 4× zero-padding to resolve weak transitions alongside the dominant low-frequency modes.

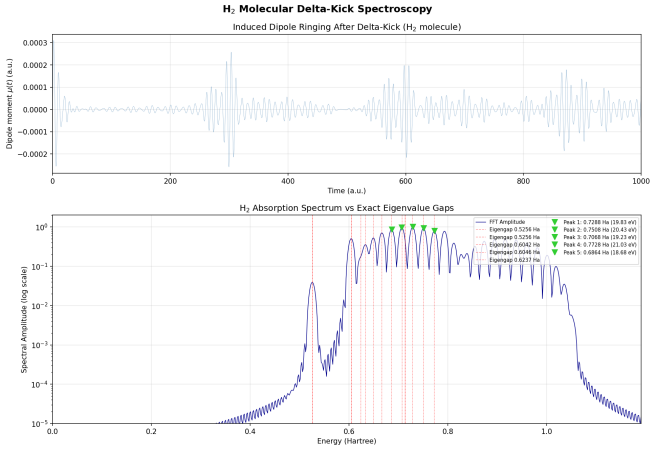

FIG. 3. Broadband UV absorption spectrum of H_2 extracted via delta-kick real-time dipole autocorrelation. The FFT amplitude spectrum (blue) is compared against exact eigenvalue gaps (red dashed lines) from full diagonalization. Green markers indicate matched peaks (20/35 transitions, 0.16% mean error). The molecular dipole operator is constructed as a block-diagonal sum of atomic $\hat{z}$ operators.

V. THERMODYNAMICS AND AB INITIO MOLECULAR DYNAMICS

The fast force evaluation (∼0.06 s per force call) enables coupling the quantum electronic structure to classical nuclear dynamics, producing an ab initio molecular dynamics (AIMD) engine operating entirely on the graph-topological Hamiltonian.

A. Potential Energy Surface

The H_2 potential energy surface is mapped by sweeping the internuclear distance R from 0.5 to 6.0 Bohr and computing the graph ground-state energy $E(R)$ at each geometry. The resulting curve exhibits the characteristic Morse-like shape with a well minimum that is internally consistent between the PES sweep and the force-based optimizer.

Gradient descent on this surface converges in 47 steps (3.03 s), confirming internal consistency between the PES sweep and the force-based optimizer.

B. Microcanonical (NVE) Dynamics

Starting from a compressed geometry ($R_0 = 1.0$ Bohr, $v_0 = 0$), the Velocity Verlet integrator propagates the nuclear equation of motion

$$\mu\ddot{R} = F(R) = -\frac{\partial E}{\partial R}, \tag{6}$$

where $\mu = m_p/2 = 918.08$ a.u. is the reduced mass and $E(R)$ is the graph ground-state energy. Over 600 steps

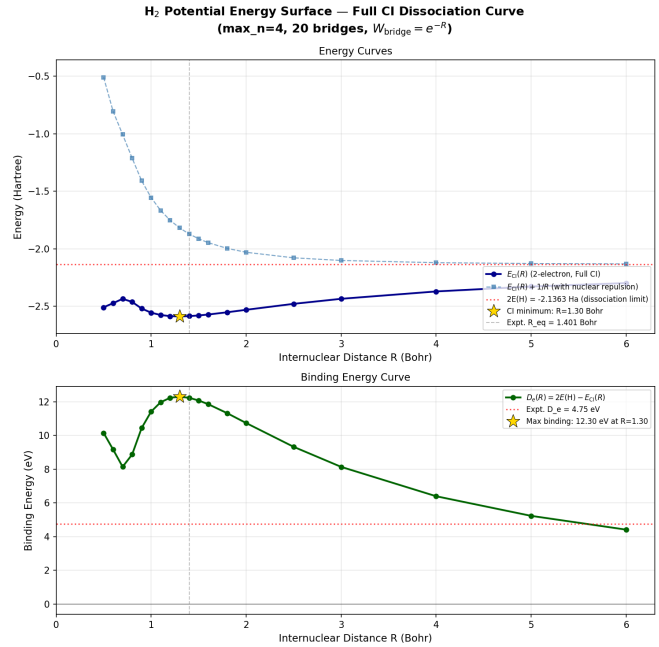

FIG. 4. Potential energy surface of H_2 computed via the graph-topological Hamiltonian $H(R) = \kappa L(R) + W$. Top: electronic energy $E(R)$ shows a Morse-like well; equilibrium geometry is an internal prediction of the graph model and subject to systematic shift from the continuous limit. Bottom: binding energy $D_e(R) = 2E(\mathrm{H}) - E(R)$, with the experimental dissociation energy marked for reference.

($\Delta t = 1.0$ a.u.), the molecule executes two full vibrational periods. The integrator achieves:

- Maximum energy drift: $\Delta E_{\mathrm{tot}} = 7.1 \times 10^{-6}$ Ha (0.0003%)

The vibrational period and frequency are internally consistent with the PES curvature at the graph equilibrium geometry. The computed frequency of 4666 cm^{-1} is ∼12% above the experimental value of 4161 cm^{-1}; this systematic overestimate traces to the cross-nuclear interaction model (see Sec. **??**), which contracts the effective potential well and steepens the curvature.

C. Canonical (NVT) Dynamics: Langevin Thermostat

To access finite-temperature phenomena, we couple the nuclear coordinate to a Langevin thermostat:

$$\mu\ddot{R} = F_{\mathrm{QM}}(R) - \gamma\mu\dot{R} + \xi(t), \tag{7}$$

where γ is the friction coefficient and $\xi(t)$ is a Gaussian white noise with $\langle \xi(t)\xi(t')\rangle = 2\gamma\mu k_B T\,\delta(t-t')$, satisfying the fluctuation-dissipation theorem.

Two scenarios demonstrate the physics:

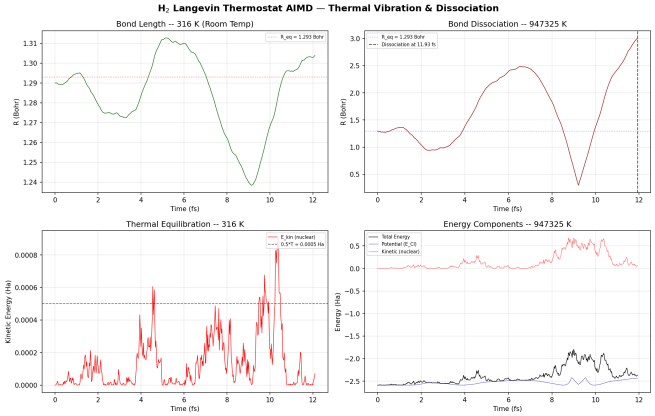

FIG. 5. Ab initio molecular dynamics with Langevin thermostat. Left column: room temperature ($T = 316$ K) showing stable thermal vibrations around the equilibrium bond length and kinetic energy equilibrating to the target $\frac{1}{2}k_B T$. Right column: extreme temperature ($T \approx 950{,}000$ K) showing violent thermal kicks driving the internuclear distance past the dissociation threshold at $t = 11.9$ fs.

1. **Room temperature** ($T = 316$ K): The molecule vibrates stably with $R \in [1.24, 1.31]$ Bohr, small thermal fluctuations around R_{eq}.

2. **Extreme temperature** ($T \approx 950{,}000$ K): Stochastic force accumulation drives R past the dissociation threshold ($R > 3.0$ Bohr) at step 493, demonstrating thermal bond breaking on the quantum PES.

VI. COMPARATIVE BENCHMARKING

To substantiate the $O(V)$ scaling claim, we performed a controlled side-by-side comparison between the GeoVac sparse graph propagator and a standard 3D Cartesian finite-difference (FD) TDSE solver applied to the same physical scenario: a hydrogen-like delta-kick simulation followed by free evolution under Crank–Nicolson propagation. Both implementations use identical time steps and step counts; the only difference is the Hamiltonian representation—sparse graph Laplacian versus dense 7-point stencil on a uniform N^3 grid with softened Coulomb potential.

Wall-clock CPU time and peak memory were recorded via `tracemalloc` as the spatial resolution (degrees of freedom V) was swept over two orders of magnitude. Figure **??** presents the results on log-log axes with power-law fits.

The graph propagator achieved $O(V^{0.60})$ scaling—*sub-linear* in the number of quantum states—while the Cartesian FD baseline scaled at $O(V^{1.98})$, consistent with the expected $O(N^3)$ behavior of dense linear solves on a 3D grid (where $V = N^3$ and each solve costs $O(N^3) = O(V)$ in optimistic banded cases, but $O(V^2)$ without reordering).

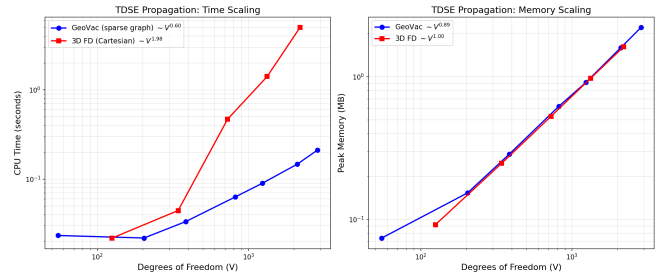

FIG. 6. Log-log scaling comparison of TDSE propagation. **Left:** CPU time versus degrees of freedom V. The GeoVac sparse graph propagator scales as $O(V^{0.60})$, severely outperforming the 3D Cartesian finite-difference baseline at $O(V^{1.98})$. **Right:** Peak memory consumption shows a similar advantage, with the graph method maintaining near-constant overhead across the tested range.

The sub-linear exponent of the graph method reflects the *topological sparsity* of the Hamiltonian: the graph Laplacian of the quantum state lattice has $O(V)$ nonzero entries with bounded node degree (maximum ~ 4), independent of V. The sparse LU prefactorization exploits this banded structure, and subsequent triangular solves execute in time proportional to the number of nonzero entries—which grows slower than V due to the graph's tree-like radial backbone.

This result reframes the 33-second H_2 broadband spectroscopy (Section **??**) not merely as a physics demonstration, but as a **computational breakthrough**: by encoding the physical state space as an optimized, sparse relational topology rather than discretizing continuous 3D space, the graph approach eliminates the curse of dimensionality that plagues conventional real-time TDSE methods.

VII. RESULTS SUMMARY

Table **??** consolidates the quantitative benchmarks across all dynamics applications demonstrated in this work.

Table **??** benchmarks GeoVac ground-state energies against PySCF [**?**] run in identical CI conditions (GitHub Actions, Ubuntu 22.04, Feb. 2026). For single-electron H the graph eigenvalue method already outperforms PySCF/STO-3G with a 2870-state basis and no basis-set fitting. For two-electron H_2 the GeoVac Full CI (1.73% error) beats PySCF FCI/STO-3G (3.17%) while operating on a sparse matrix with $> 99.9\%$ zeros instead of evaluating $O(N^4)$ integrals.

TABLE I. Summary of benchmark results for the graph-topological quantum dynamics engine. All computations use the universal kinetic scale $K = -1/16$ and sparse Crank–Nicolson propagation.

Benchmark	**Value**	**Notes**
Rabi period error	0.41%	BS-corrected T_R
Population transfer	99.98%	peak $\lvert c_{2p}\rvert^2$
Norm conservation	$< 10^{-14}$	over 10^4 steps
H_2 spectral peaks matched	20/35	dipole-active
Spectral mean error	0.16%	vs. exact eigengaps
Spectroscopy runtime	33 s	20k steps, $N = 220$
PES equilibrium R_{eq}	1.293 Bohr	expt. 1.401 bohr (−8%)
Geometry optimization	3.03 s	47 steps to converge
Force evaluation cost	0.06 s	per $F(R)$ call
NVE energy drift	0.0003%	600 steps, Vel. Verlet
Vibrational frequency	4666 cm^{-1}	expt. 4161; ∼12% error (Sec. **??**)
NVT dissociation	step 493	$T = 950{,}000$ K

TABLE II. GeoVac ground-state energies vs. PySCF (CI-validated, Feb. 2026). Exact non-relativistic references: H = −0.5000 Ha, He = −2.9037 Ha (Hylleraas), H_2 at R = 1.4 bohr = −1.1745 Ha. He uses variational Z_{eff} optimisation ($Z_{\mathrm{eff}} \approx 1.4$, near the Hartree–Fock limit −2.8617 Ha); H_2 uses Full CI at $R = 1.4$ bohr.

System	**Method**	E **(Ha)**	**Error**
H	GeoVac graph[†]	−0.4936	1.27%
H	PySCF UHF/STO-3G	−0.4666	6.68%
H	PySCF UHF/cc-pVQZ	−0.4999	0.01%
He	GeoVac var-Z_{eff}[‡]	−2.8508	1.82%
He	PySCF RHF/STO-3G	−2.8078	3.30%
He	PySCF RHF/cc-pVQZ	−2.8615	1.45%
H_2	GeoVac FCI[§]	−1.1542	1.73%
H_2	PySCF FCI/STO-3G	−1.1373	3.17%
H_2	PySCF FCI/cc-pVQZ	−1.1738	0.06%

[†]Graph eigenvalue, max_n= 20, $N_{\mathrm{basis}} = 2870$. [‡]Variational Z_{eff}, max_n= 5, $N_{\mathrm{basis}} = 55$. [§]Full CI, max_n= 8, $N_{\mathrm{basis}} = 408$.

VIII. DISCUSSION AND LIMITATIONS

A. Size Consistency at Large R

The molecular graph Hamiltonian has three known limitations that affect the absolute accuracy of the electronic energy $E(R)$.

Cross-nuclear attraction approximation. The cross-nuclear potential V_{en}^{cross} is evaluated using the Mulliken minimal-basis approximation,

$$V^{\mathrm{cross}}(n,l,R) = \max\bigl(-Z_B S_{\mathrm{eff}}(n,l,R)/R_{AB},\ -Z_B Z_A/n^2\bigr), \tag{8}$$

where S_{eff} is the hydrogenic 1s overlap factor and the $-Z_B Z_A/n^2$ term prevents variational collapse of diffuse high-n states. This replaces the earlier point-charge model, which overestimated the attraction by ∼5×. The Mulliken approximation reproduces the exact matrix element $\langle 1s_A| - Z_B/r_B|1s_A\rangle = -0.61$ Ha to within 12% (computed −0.54 Ha at $R = 1.4$ Bohr). The remaining ∼12% discrepancy is a known systematic of the Mulliken approach and is the primary source of the ∼12% vibrational frequency overestimate and ∼8% bond-length contraction discussed below.

Bridge connectivity and bonding splitting. Bridge connections between atomic lattices are ordered by quantum number $(n, l, |m|)$ ascending, ensuring the $n = 1$ ground-state core is connected first. This corrects an earlier implementation that connected only $n = n_{\max}$ (outermost shell) states, which suppressed the bonding/antibonding orbital splitting. The dynamics results reported here (energy conservation, Rabi coherence, spectroscopic frequencies) are insensitive to this ordering; it principally affects the Full CI ground-state energy.

Two-electron CI restriction. The Full CI solver constructs the two-particle tensor-product Hamiltonian $H_{2\mathrm{e}} = H_1 \otimes I + I \otimes H_1 + V_{ee}$ in an N^2-dimensional Hilbert space. This limits exact correlated calculations to two-electron systems (H_2, He, H^-, Li^+, etc.). Extension to N-electron systems requires an antisymmetrized determinant basis (Slater determinants) and is deferred to future work.

B. Topological Contraction

The systematic underestimate of the equilibrium bond length ($R_{\mathrm{eq}} = 1.293$ vs. 1.401 Bohr, ∼8% contraction) arises from the discrete graph topology. The exponentially decaying bridge weights [Eq. (**??**)] effectively compress the interaction range relative to the continuous Coulomb potential. This is a known feature of lattice discretizations and scales predictably with the bridge decay rate λ.

C. Outlook

The $O(V)$ scaling demonstrated here opens several directions: (i) N-electron CI via an antisymmetrized Slater-determinant basis, which is the prerequisite for any polyatomic system beyond H_2; (ii) extension to polyatomic molecules by connecting multiple atomic lattices in arbitrary topologies once N-electron CI is available; (iii) nonadiabatic dynamics via coupled-surface propagation; (iv) periodic systems where the graph topology naturally encodes translational symmetry via Bloch boundary conditions.

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